COMPATIBLE SYSTEMS OF NONEMPTY CLOSED SUBSETS HAVE A POINT IN THE LIMIT

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Limit nonempty. Limits, Lemma lemma-limit-nonempty (same chapter)
- Affine permanence. Morphisms, Lemma morphisms-lemma-affine-permanence

Source reference. Original reference 086P: sources/stacks-chow/tags.txt, source lines 10399–10399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SE: sources/stacks-chow/tags.txt, source lines 2335–2335. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SC: sources/stacks-chow/tags.txt, source lines 2333–2333. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SG: sources/stacks-chow/tags.txt, source lines 2337–2337. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z2: sources/stacks-chow/tags.txt, source lines 2577–2577. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 086P. Suppose for each i we are given a nonempty closed subset $Z_i \subset S_i$ with $f_{i'i}(Z_{i'}) \subset Z_i$ for all $i' \geq i$. Then there exists a point $s \in S$ with $f_i(s) \in Z_i$ for all i.*

Proof. Let $Z_i \subset S_i$ also denote the reduced closed subscheme associated to Z_i, see Schemes, Definition 01J4. A closed immersion is affine, and a composition of affine morphisms is affine (see Morphisms, Lemmas 01SE and 01SC), and hence $Z_{i'} \to S_i$ is affine when $i' \geq i$. We conclude that the morphism $f_{i'i} : Z_{i'} \to Z_i$ is affine by Morphisms, Lemma 01SG. Each of the schemes Z_i is quasi-compact as a closed subscheme of a quasi-compact scheme. Hence we may apply Lemma 01Z2 to see that $Z = \lim_i Z_i$ is nonempty. Since there is a canonical morphism $Z \to S$ we win. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.